Exp 5 · BJT input/output characteristics (CE) and h-parameters

Manual pp. 17–19 · LTspice / Virtual Lab · CO1, CO2 · theory: ECE2101 unit 5 (BJTs)

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What you already have: a diode is one pn junction (Exp 1). It conducts forward above ≈ 0.7 V.

Three terms first:

- **BJT (bipolar junction transistor):** three layers, **n-p-n** for an NPN. The layers are the **emitter** (E, heavily doped), the **base** (B, very thin, lightly doped) and the **collector** (C). That makes two junctions: base–emitter (BE) and base–collector (BC).
- **Common emitter (CE):** the emitter is shared by the input (B–E) and the output (C–E).
- **h-parameters:** four numbers that describe how the transistor responds to small changes. Exp 6–8 amplifiers are built on them.

MANUAL EXP 5, AIM

To plot the input and output characteristics of an NPN transistor in CE configuration and determine h-parameters from the characteristics. Viva form: "Your output curves are flat lines spaced 5.5 mA apart for 20 µA steps of base current. What is h_{fe}, and what does a flat line tell you?"

GUESS FIRST

Commit to a number for h_{fe}, and to a one-word meaning of "flat", before reading on.

The BJT in five lines

1. Forward-bias the BE junction (V_{BE} ≈ 0.6–0.7 V). The emitter injects electrons into the base.
2. The base is thin, so almost all of them cross it and are swept into the collector by the reverse-biased BC junction. A few recombine in the base: that is the base current.
3. So a **small base current controls a large collector current**: $I_C = \beta I_B$, with β ≈ 100–300 (≈ 290 for the BC547B).
4. KCL: $\mathbf{I_E = I_B + I_C}$.
5. Which region the transistor is in depends on how the two junctions are biased:

Region	BE junction	BC junction	Behaviour	Used as
Active	forward	reverse	$I_C = \beta I_B$, nearly independent of V_{CE}	amplifier (Exps 6–8, 11)
Saturation	forward	forward	V_{CE} ≈ 0.2 V, $I_C < \beta I_B$	closed switch
Cut-off	reverse	reverse	I_C ≈ 0	open switch

The two families of curves

- **Input characteristic:** I_B against V_{BE}, at fixed V_{CE}. It is just the BE diode's forward curve.
- **Output characteristic:** I_C against V_{CE}, at fixed I_B. One curve per I_B.

h-parameters in CE form (small changes, written in lowercase):

$$v_{be} = h_{ie}\, i_b + h_{re}\, v_{ce} \qquad i_c = h_{fe}\, i_b + h_{oe}\, v_{ce}$$

Parameter	Name	Read from	Formula
h_{ie} (= h_{11})	input resistance	input curve, V_{CE} fixed	$\Delta V_{BE}/\Delta I_B$
h_{re} (= h_{12})	reverse voltage gain	input curves, I_B fixed	$\Delta V_{BE}/\Delta V_{CE}$
h_{fe} (= h_{21})	forward current gain	output curves, V_{CE} fixed	$\Delta I_C/\Delta I_B$
h_{oe} (= h_{22})	output admittance	one output curve, I_B fixed	$\Delta I_C/\Delta V_{CE}$

The manual swaps the names for h_{21} and h_{22}, and writes "$1/h_{oe} = \Delta I_C/\Delta V_{CE}$". The table above is the correct version. $\Delta I/\Delta V$ is a conductance, so h_{oe} is in siemens and $1/h_{oe}$ is the output resistance.

The circuit, and one fix

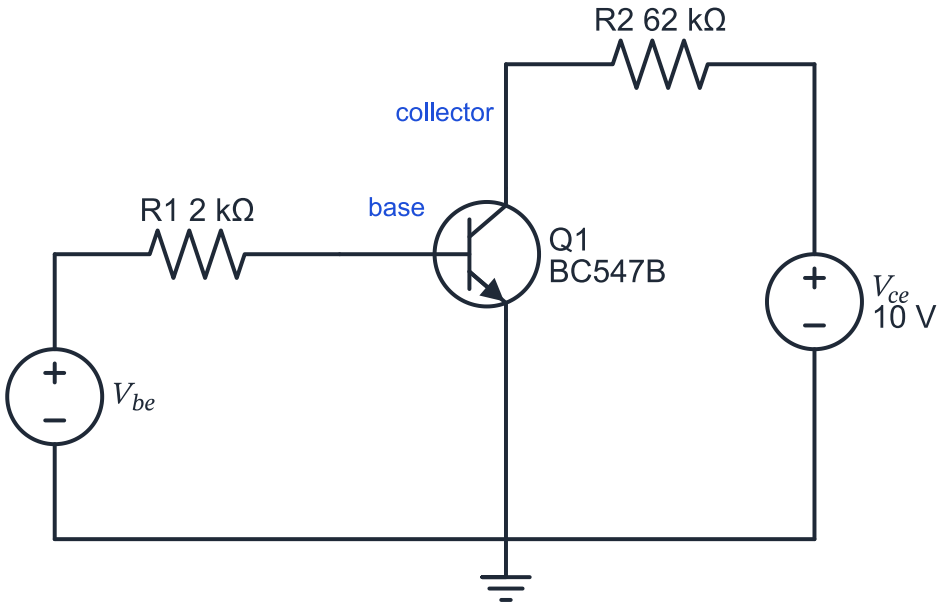


Fig 5.1 as printed: V_{be} through 2 kΩ to the base; V_{ce} through 62 kΩ to the collector.

TRAP: THE 62 KΩ MAKES THE MANUAL'S TABLE IMPOSSIBLE

With 62 kΩ in the collector, $I_C \le 10\ V / 62\ k\Omega =$ **0.16 mA**. The transistor saturates as soon as I_B passes about 0.6 µA, so "V_{CE}" is no longer the 10 V you set, and the I_B = 20/40 µA curves of the observation table can never appear. That is why the manual's own output plot only reaches about 210 µA. The fix: **replace R2 with a wire** so V_{ce} sits directly on the collector, and drive the base with a **current source** for the output curves. The device and the directives stay the same.

DO IT (LTSPICE)

Input characteristics:

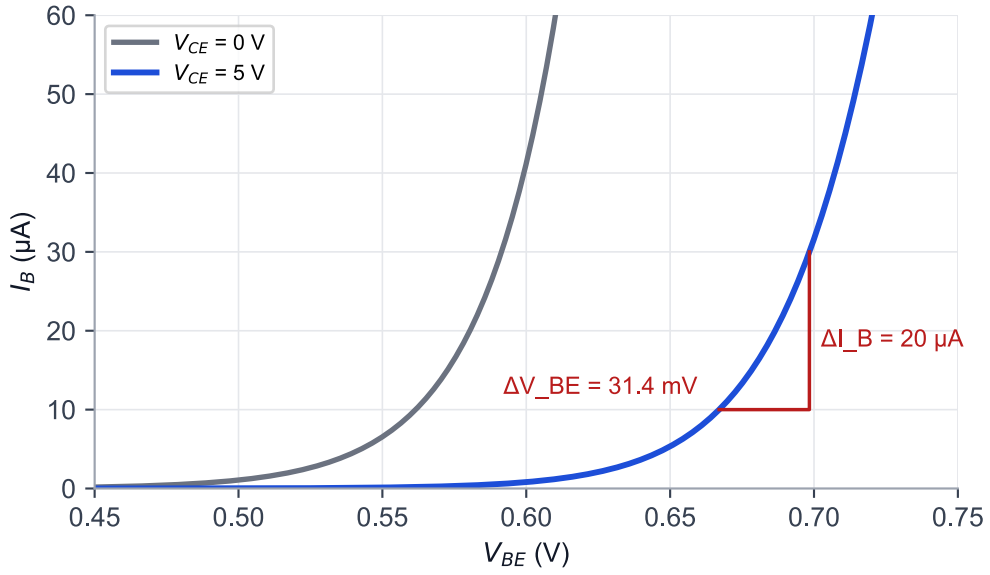
1. Draw Fig 5.1 with R2 replaced by a wire. Q1: Pick New Transistor → **BC547B**. Label the base node `base`.
2. Directive: `.dc Vbe 0 1 0.001 Vce 0 5 5` (V_{CE} = 0 V and 5 V, as the observation table asks).
3. Run, click the base lead to plot **Ib(Q1)**, then right-click the x-axis and set it to **V(base)**.

Output characteristics:

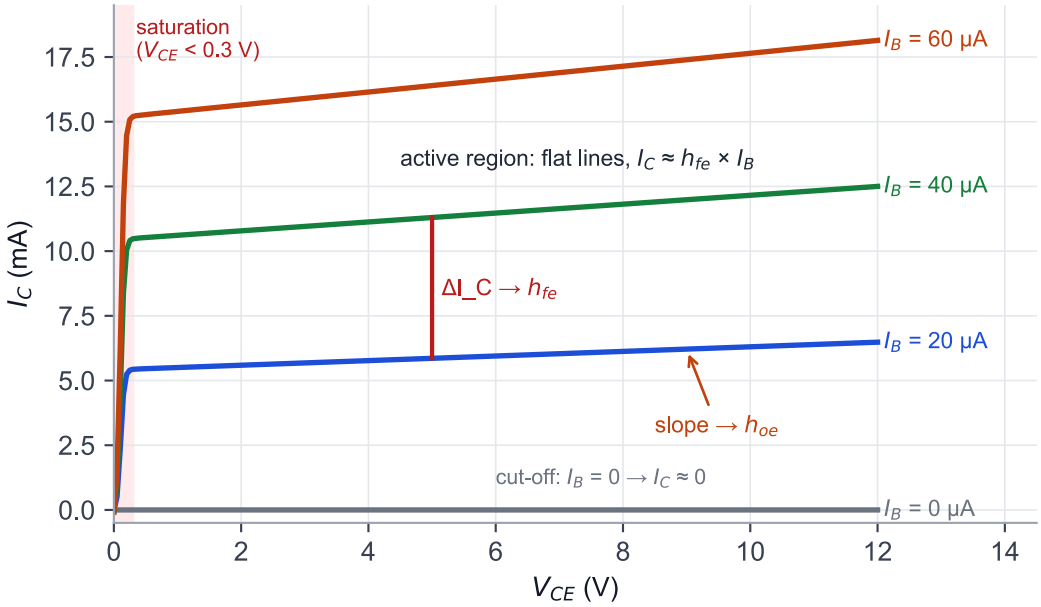
1. Replace V_{be} and R1 with a **current source** `Ib` (arrow pointing **into** the base).
2. Directive: `.dc Vce 0 12 0.05 Ib 0 60u 20u` (I_B = 0, 20, 40, 60 µA).
3. Plot **Ic(Q1)** against V_{ce}.

h_{re}: set Ib = `20u`, run `.op` at V_{ce} = 2 V and at 10 V, and read V(base) both times.

What you should see



Input characteristic: a diode curve. Raising V_{CE} shifts it right (at V_{CE} = 0 the BC junction is also forward and adds base current).



Output characteristics: saturation hugging the left edge, the active region (nearly flat), and cut-off at $I_B = 0$.

V_{CE} (V)	0.5	1	2	4	6	8	10	12
I_C (mA), I_B = 0	0	0	0	0	0	0	0	0
I_C (mA), I_B = 20 µA	5.46	5.50	5.59	**5.77**	5.95	6.13	**6.30**	6.48
I_C (mA), I_B = 40 µA	10.53	10.61	10.78	11.13	11.47	11.81	12.16	12.50
I_C (mA), I_B = 60 µA	15.27	15.40	15.65	16.15	16.65	17.14	17.64	18.14

Input, V_{CE} = 5 V: V_{BE} = 0.60 V → 0.81 µA; 0.62 → 1.73; 0.64 → 3.68; 0.66 → 7.73; 0.68 → 15.9; 0.70 → 31.6 µA. At V_{CE} = 0 V: 0.50 V → 1.07 µA; 0.55 → 6.55; 0.60 → 41.2 µA.

Calculations (from the curves above)

$$h_{ie} = \frac{\Delta V_{BE}}{\Delta I_B} = \frac{0.6984 - 0.6670}{30 - 10\ \mu A} = \frac{31.4\ mV}{20\ \mu A} = 1.57\ k\Omega \qquad (V_{CE} = 5\ V)$$

$$h_{fe} = \frac{\Delta I_C}{\Delta I_B} = \frac{11.30 - 5.86\ mA}{40 - 20\ \mu A} = \frac{5.44\ mA}{20\ \mu A} = 272 \qquad (V_{CE} = 5\ V)$$

$$h_{oe} = \frac{\Delta I_C}{\Delta V_{CE}} = \frac{6.30 - 5.77\ mA}{10 - 4\ V} = \frac{0.533\ mA}{6\ V} = 89\ \mu S \qquad (I_B = 20\ \mu A;\ 1/h_{oe} = 11.2\ k\Omega)$$

$$h_{re} = \frac{\Delta V_{BE}}{\Delta V_{CE}} = \frac{0.68671 - 0.68638}{10 - 2} = \frac{0.33\ mV}{8\ V} = 4.2 \times 10^{-5} \qquad (I_B = 20\ \mu A)$$

Sense check: h_{re} is tiny (the output barely affects the input), and h_{oe} is small (the flat lines tilt only slightly). That is why simple amplifier analysis often drops both.

WRITE IN YOUR RECORD

Result (BC547B, CE): $h_{ie} \approx$ **1.57 kΩ**, $h_{re} \approx$ **4.2×10^{-5}**, $h_{oe} \approx$ **89 µS**, $h_{fe} \approx$ **272**. State the operating point used: V_{CE} = 5 V, $I_B \approx$ 20 µA.

Theory for the record: a BJT has emitter, base and collector regions forming two junctions. For amplification the BE junction is forward-biased and the BC junction reverse-biased (active region): carriers injected from the emitter cross the thin base and reach the collector, so $I_C = \beta I_B$ and $I_E = I_B + I_C$. In CE configuration, the input characteristic (I_B vs V_{BE} at constant V_{CE}) resembles a forward diode curve. The output characteristics (I_C vs V_{CE} at constant I_B) show saturation, active and cut-off regions. The h-parameters h_{ie}, h_{re}, h_{fe}, h_{oe} are the slopes of these curves at the operating point and form the small-signal model.

CHECK 1

(a) Using the table, find h_{fe} at V_{CE} = 10 V between I_B = 40 and 60 µA. (b) Why is the output curve for I_B = 0 flat at zero? (c) Which region is the transistor in at V_{CE} = 0.1 V and I_B = 40 µA, and how can you tell?

Answers

Opening question. h_{fe} = 5.5 mA / 20 µA ≈ **275**. "Flat" means **I_C does not depend on V_{CE}**: the transistor acts as a current source controlled by I_B. That is the active region, and the slight tilt is h_{oe}.

Check 1. (a) (17.64 − 12.16) mA / 20 µA = 5.48 mA / 20 µA = **274**. (b) With no base current, the BE junction isn't injecting carriers, so the transistor is cut off and only a leakage current (nA) flows. (c) **Saturation.** V_{CE} is below about 0.2–0.3 V, the steep left-hand part of the curve, where I_C is less than βI_B and depends strongly on V_{CE}.